

Automated Always-On Poultry Incubator

In a country like India where blackouts lasting up to eight hours a day are frequent in rural areas, an automated battery-powered poultry incubator is quite the life saver for poultry farmers. The best part is it can bring down their power consumption by a whopping 50 per cent!



Electronics setup for the incubator

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Poultry incubators need a constant source of power to maintain optimal temperature, humidity and other parameters for successful egg hatching. However, in India, power failures are a major headache for the poultry farmers. Some of them have diesel generators to offset the effect, but this drives the cost of eggs up and small-scale farmers get affected the most.

Karthik K.G. and Sakthi Prakash P., managing partners at Pluto India, have come up with an automated battery-run poultry incubator to counter this problem.

What's an incubator

An incubator is a device that simulates avian incubation by keeping eggs warm and at the correct humidity, to hatch them. Modern incubators are electrically heated with a thermostat.

The incubator maintains optimal temperature, humidity and air quality (CO_2) inside in the total 21 days of incubation. In the first



Electronics setup connected to the incubator

18 days, eggs are placed in a chamber called 'setter.' Here the eggs need to be tilted frequently during the day in order to avoid chicks getting stuck to the egg shell. During the remaining three days, they are kept in a chamber called 'hatcher.'

Whenever the CO_2 level inside the chamber reaches an alarming value, the exhaust fan runs to fill the chamber with fresh air. To maintain temperature and humidity, a heater and an ultrasonic humidifier, respectively, are used.

The solution

The battery-run incubator developed by Karthik and Sakthi uses a 150Ah battery and inverter set-up. While designing a power-efficient solution at every level, they have also ensured the quality of egg hatching. During the first week, they had a pretty low mortality rate for the chicks.

The system is automated, sensing all the parameters associated with a control cycle and logging their status onto a computer. This data includes incubator/hatcher door status, sensor-1 humidity, sensor-2 humidity (redundancy), sensor-1 temperature, sensor-2 temperature (redundancy), CO_2